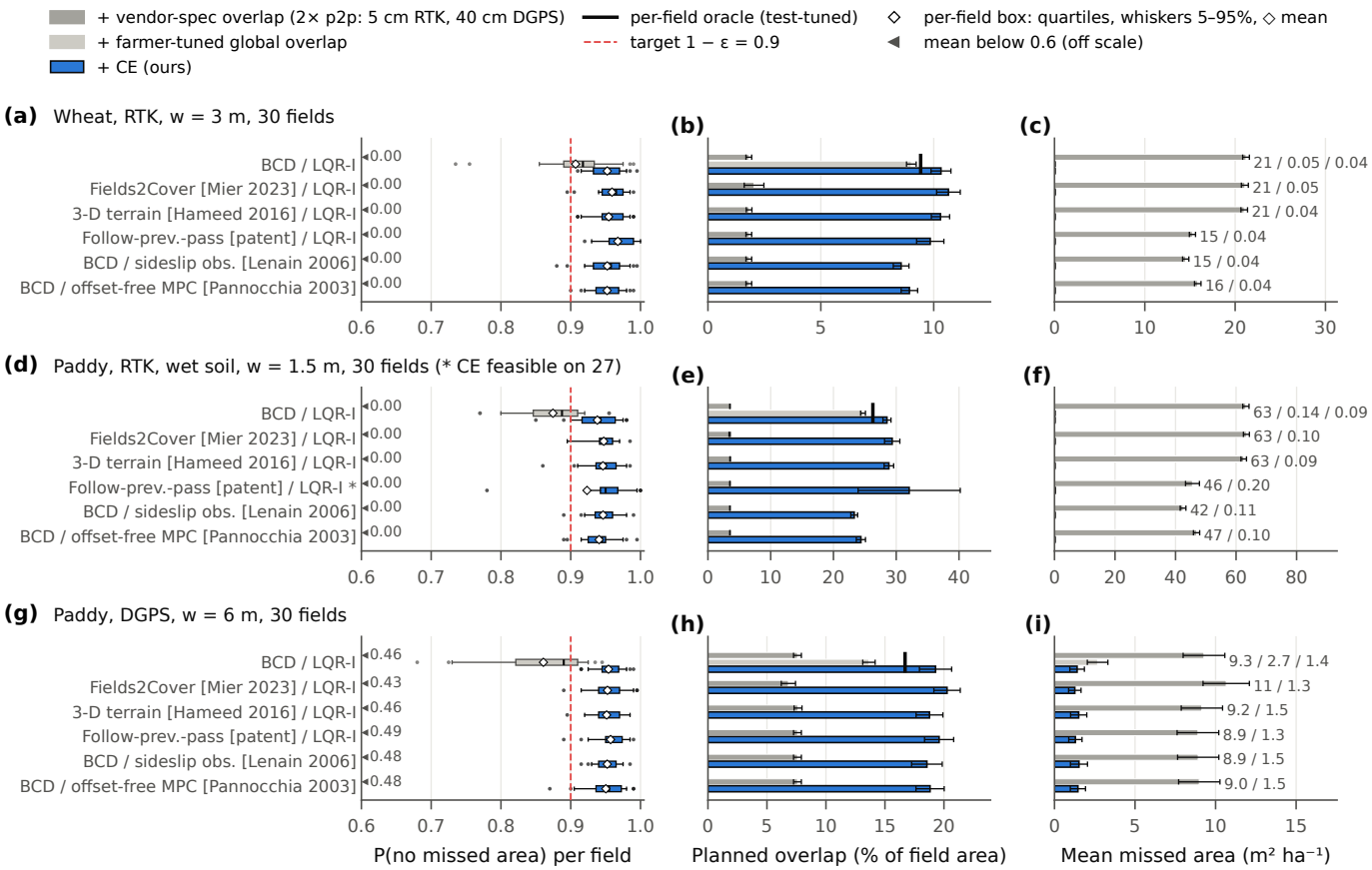
+ vendor-spec overlap (2× p2p: 5 cm RTK, 40 cm DGPS)
+ farmer-tuned global overlap
+ CE (ours)
per-field oracle (test-tuned)
target 1 − ε = 0.9
per-field box: quartiles, whiskers 5–95%, ◇ mean
mean below 0.6 (off scale)
(a) Wheat, RTK, w = 3 m, 30 fields
BCD / LQR-I
Fields2Cover [Mier 2023] / LQR-I
3-D terrain [Hameed 2016] / LQR-I
Follow-prev.-pass [patent] / LQR-I
BCD / sideslip obs. [Lenain 2006]
BCD / offset-free MPC [Pannocchia 2003]
0.00
0.6 0.7 0.8 0.9 1.0
(b)
0 5 10
(c)
21 / 0.05 / 0.04
21 / 0.05
21 / 0.04
15 / 0.04
15 / 0.04
16 / 0.04
0 10 20 30
(d) Paddy, RTK, wet soil, w = 1.5 m, 30 fields (* CE feasible on 27)
Follow-prev.-pass [patent] / LQR-I *
0.6 0.7 0.8 0.9 1.0
(e)
0 10 20 30 40
(f)
63 / 0.14 / 0.09
63 / 0.10
63 / 0.09
46 / 0.20
42 / 0.11
47 / 0.10
0 20 40 60 80
(g) Paddy, DGPS, w = 6 m, 30 fields
0.46
0.43
0.46
0.49
0.48
0.48
0.6 0.7 0.8 0.9 1.0
P(no missed area) per field
(h)
0 5 10 15 20
Planned overlap (% of field area)
(i)
9.3 / 2.7 / 1.4
11 / 1.3
9.2 / 1.5
8.9 / 1.3
8.9 / 1.5
9.0 / 1.5
0 5 10 15
Mean missed area (m² ha⁻¹)